

Assignment 2

1. Assignment Description

Problem: Establishment of the Set Class (50 points)

The Pascal language provides built-in operations for set classes, while the C language does not. In this assignment, you are required to implement a set class named `TSet` using C++. This class will facilitate mathematical operations on sets, providing functionality for union, intersection, difference, containment, and belonging.

Set Operations and Symbols

Your `TSet` class must use **operator overloading** to implement the following symbols.

Assume A and B are both sets. The following symbols are used to define each operation:

- Union ($A + B$): Returns the union of set A and set B .
- Intersection ($A * B$): Returns the intersection of set A and set B .
- Difference ($A - B$): Returns the difference of set A and set B .
- Containment ($A \geq B$): Returns `true` if set A contains set B .
- Belonging ($x \text{ in } A$): Returns `true` if character x belongs to set A .

Input

- The first line contains an integer N , representing the number of test cases.
- Each test case consists of three lines:
 1. Elements of set A : The first integer indicates the number of elements in the set, followed by the elements themselves.
 2. Elements of set B : The first integer indicates the number of elements in the set, followed by the elements themselves.
 3. A single integer x to be tested for belonging.
- Note:
 - Each input element in a line is separated by a comma.
 - The integers in one line belong to the same set (excluding the first integer, which represents the size of the set).
 - The input integers are not ordered.
 - There may be some duplicated integers, but the same duplicated integers in a set can be counted only once.
- Constraints
 - number of test cases N : $1 \leq N \leq 20$.
 - The sizes of sets A and B : $0 \leq |A|, |B| \leq 10,000$.
 - The values of the elements: $0 \leq A_i, B_i, x \leq 255$. ($0 \leq A_i, B_i, x \leq 2^{32} - 1$ for the bonus test cases)

Output

- First, print the case number as "Test Case #X".
- Print the contents of sets A and B (elements must be sorted in ascending order, duplicates removed).
- Print the results of the five operations: $A+B$, $A*B$, $A-B$, $B-A$, containment, and belonging checks.
- Print a blank line between two test cases.

Sample Input

```
2
6,1,2,3,4,5,6
8,3,8,6,5,5,3,8,9
8
7,3,10,11,15,4,14,12
2,4,3
17
```

Sample Output

```
Test Case 1:
A:1,2,3,4,5,6
B:3,5,6,8,9
A+B:1,2,3,4,5,6,8,9
A*B:3,5,6
A-B:1,2,4
B-A:8,9
A does not contain B
B does not contain A
8 is not in A
8 is in B

Test Case 2:
A:3,4,10,11,12,14,15
B:3,4
A+B:3,4,10,11,12,14,15
A*B:3,4
A-B:10,11,12,14,15
B-A:
A contains B
B does not contain A
17 is not in A
17 is not in B
```

2. Submission Guidelines

Please organize your code and rename your files according to the following format before uploading.

- **Source Code Archive (.zip):** Place your main function code, all custom .cpp and .h files, and any other files required for compilation into a specific folder. Compress this folder into a .zip file. (Note: If you organize your source codes into multiple subdirectories, you must include a `Makefile` in the root of your folder.)
- **Compiled Executable File (.exe):** Submit the compiled executable file alongside your .zip archive.

- Naming example
 - Folder Name: 1142DS_<Student ID>_a2
 - Zip File: 1142DS_<Student ID>_a2.zip (e.g., 1142DS_B11234567_a2.zip)
 - Executable: 1142DS_<Student ID>_a2.exe (e.g., 1142DS_B11234567_a2.exe)

3. Grading Policies

General Criteria (15 points):

- Coding Style (10 points): Assessed based on code readability, consistent indentation, and meaningful naming conventions. Arbitrary or meaningless variable names (e.g., a, b, c, aa, bb, cc) are strictly prohibited and will result in point deductions.
- Comments (5 points): Assessed based on brief and clear explanations of your function purposes and the overall processing flow. Focus on explaining the core logic. Do not add trivial or redundant comments for basic C++ syntax (e.g., commenting on `#include` or `using namespace std`). Additionally, every source file must include a file header specifying the author (including student ID), the date of writing (e.g., Mar. 22, 2026), and a description of the program's functionality and usage.

Problem-Specific Criteria (35 points)

- Correctness (30 points): Points are awarded based on the percentage of test cases your program successfully passes. The default test cases are based on character sets, where each element corresponds to an ASCII value (0–255). Therefore, a fixed-size array of 256 is sufficient for the basic requirement. If you do **not use operator overloading**, your program will receive **zero points**, even if the functionality is correct.
- Bonus (5 points): Additional test cases may use integer sets with elements ranging from 0 to $2^{32}-1$. Since these sets are much larger than the ASCII range, a suitable data structure should be used instead of a fixed-size array.

4. Restrictions (Not Allowed)

Unless explicitly permitted by the assignment prompt, the usage of container libraries in the C++ Standard Template Library (STL)—such as `vector`, `list`, `map`, `set`, `deque`, or similar container structures—is strictly prohibited. If such libraries are used, the correctness score will be automatically **marked as 0**.